📘 AI/ML Internship – Day 30 Notes & Tasks

🟢 Module 6: Data Visualization Basics with Matplotlib

📌 Part 1: What is Data Visualization?

🔹 Definition

👉 Data visualization means representing data using:

- Charts
- Graphs
- Plots

👉 Instead of reading large tables, graphs make data easier to understand.

📌 Part 2: Why Visualization is Important?

👉 In AI/ML:

- Helps identify patterns
- Detect trends
- Understand datasets visually

👉 Example:

- Sales growth
- Student performance
- Temperature changes

📌 Part 3: Theory Answers (IMPORTANT)

🔹 1. What is Matplotlib?

👉 Matplotlib is a Python library used for creating graphs and charts.

🔹 2. Why Visualization is Important?

👉 Visualization:

- Makes data easy to understand
- Helps analyze patterns
- Improves decision making

🔹 3. What is a Plot?

👉 A plot is a graphical representation of data.

🔹 4. What is X-axis and Y-axis?

Axis	Meaning
X-axis	Horizontal axis
Y-axis	Vertical axis

🔹 5. Why Graphs are Used in AI?

👉 Graphs help visualize:

- Trends
- Accuracy
- Predictions
- Model performance

📌 Part 4: Installing Matplotlib

```
pip install matplotlib
```

📌 Part 5: Import Matplotlib

```
import matplotlib.pyplot as plt
```

📌 Part 6: Simple Line Plot

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [10, 20, 30, 40]

plt.plot(x, y)

plt.show()
```

📌 Part 7: Adding Labels & Title

```
plt.plot(x, y)

plt.title("Student Marks")
plt.xlabel("Subjects")
plt.ylabel("Marks")

plt.show()
```

📌 Part 8: Changing Line Style

```
plt.plot(x, y, linestyle="--", marker="o")

plt.show()
```

📌 Part 9: Bar Chart

🔹 Used for Comparison

```
students = ["A", "B", "C"]
marks = [80, 90, 75]

plt.bar(students, marks)
```

```
plt.show()
```

📌 Part 10: Pie Chart

🔹 Used for Percentage Representation

```
data = [40, 30, 20, 10]

labels = ["AI", "Web", "Cloud", "Cyber"]

plt.pie(data, labels=labels)

plt.show()
```

📌 Part 11: Scatter Plot

🔹 Used for Relationship Analysis

```
x = [1, 2, 3, 4]
y = [10, 15, 20, 25]

plt.scatter(x, y)

plt.show()
```

📌 Part 12: Grid & Legend

```
plt.plot(x, y, label="Marks")

plt.legend()
plt.grid()

plt.show()
```

📌 Part 13: Real-World Thinking

👉 In AI/Data Science:

- Accuracy graphs
- Loss graphs
- Sales analysis
- Weather visualization

👉 Visualization helps explain results clearly.

📌 Part 14: Visualization Example

Marks Graph

```
40 |                *
30 |             *
20 |          *
10 |       *
   +------------------------
     1   2   3   4
```

🎯 Tasks for Day 30

🧠 Task 1: Theory (Write Answers)

1. What is Matplotlib?
2. Why visualization is important?
3. What is plot?
4. Difference between X-axis and Y-axis
5. Why graphs are important in AI?

💻 Task 2: Line Plot Practice

1. Create simple line graph

2. Add labels
3. Add title

💻 Task 3: Styling Graphs

1. Add marker
2. Change line style
3. Add grid

💻 Task 4: Bar Chart Practice

1. Student marks comparison
2. Sales comparison
3. Product comparison

💻 Task 5: Pie Chart Practice

1. Department distribution
2. Expense analysis
3. Course popularity

💻 Task 6: Scatter Plot Practice

1. Height vs weight
2. Study hours vs marks
3. Temperature vs sales